

AI Deception Analyzer

Overview

The **AI Deception Analyzer** is a machine learning-based web application designed to analyze text input and predict whether it is likely to be truthful or deceptive. It also identifies the emotional tone of the text and presents the results through a simple and visually interactive interface.

This project demonstrates the practical application of **Natural Language Processing (NLP)** and **Machine Learning** in understanding human communication patterns.

Objective

The main objective of this project is to:

- Analyze user-provided text
 - Predict the probability of truthfulness vs deception
 - Detect the emotional tone of the text
 - Present results through a modern and user-friendly interface
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Problem Statement

In today's digital world, people communicate mostly through text messages, emails, and online platforms. However, text communication often lacks tone, making it difficult to understand the true intention behind a message.

This project aims to address this by using AI to analyze linguistic patterns and provide insights into whether a statement is likely truthful or deceptive.

Technologies Used

◆ **Backend**

- Python
- FastAPI
- scikit-learn
- pandas
- numpy
- joblib

◆ **Frontend**

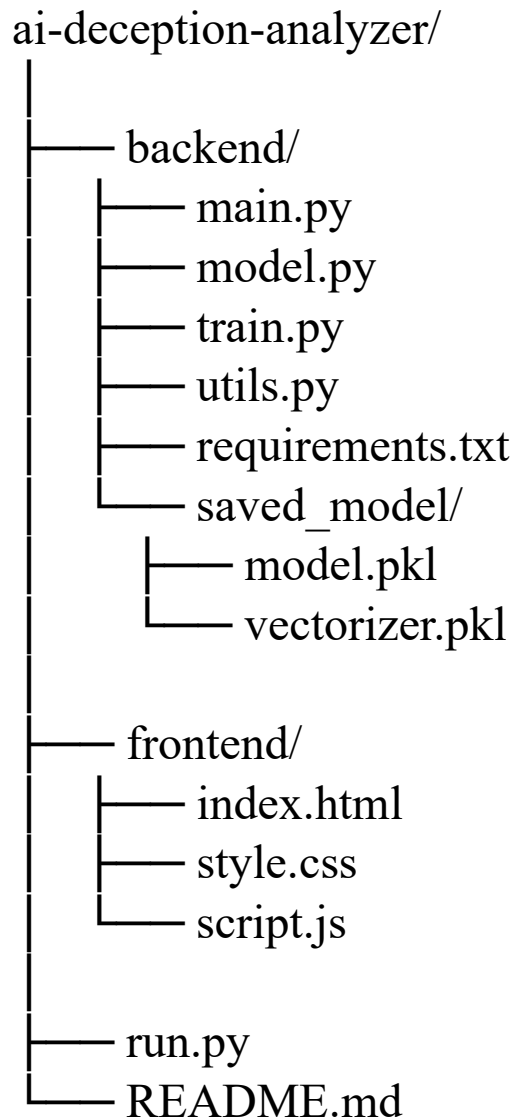
- HTML
 - CSS
 - JavaScript
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Features

-  Deception detection using Machine Learning
-  Emotion analysis (Positive, Negative, Neutral, Angry)
-  Real-time predictions via API
-  Clean and modern gradient-based UI

-  Interactive result visualization
-  Fully executable from command line

Project Structure



How It Works

1. The user enters text in the frontend
2. The text is sent to the FastAPI backend

3. The backend processes the text using a trained ML model
 4. The model predicts:
 - Truth probability
 - Deception probability
 5. Emotion is detected using keyword-based logic
 6. The results are displayed visually in the frontend
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Setup Instructions

1 Clone the Repository

```
git clone https://github.com/codeliferitesh/ai-deception-analyzer.git
cd ai-deception-analyzer
```

2 Install Dependencies

```
cd backend
pip install -r requirements.txt
```

3 Train the Model

```
python train.py
```

4 Run the Project

```
python run.py
```

Usage

- Open the application in your browser
 - Enter any text or sentence
 - Click on **Analyze**
 - View:
 - Truth score
 - Deception score
 - Emotion
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Example Input

I swear I did not do anything wrong.

Expected Output

- Truth: Low
 - Deception: High
 - Emotion: Neutral
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Future Improvements

- Integration of deep learning models like BERT
- Training on larger real-world datasets
- Improved emotion detection using advanced NLP
- Graph-based visualization of results
- User authentication and history tracking



Author

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Acknowledgement

This project has been developed as part of the **Fundamentals of AI and ML evaluated course (BYOP)**. It demonstrates the practical application of machine learning and natural language processing in solving real-world problems.



Note

If the model files are missing or corrupted, they will be automatically generated after running the training script.



Footer

Made with  by Ritesh Kumar Verma
